

1A Study Guide: Academic and Personal Skills

Learning target 1A

The development of academic and personal skills occurs through deliberate practice.

How to use this guide

You don't need the textbook to use this guide. Every textbook page that matters for 1A is copied into **Part 1**, in order, with the important sentences highlighted. Each highlight has a numbered badge that matches a note in the column beside the page. The notes explain hard words, answer the textbook's questions, and connect the page to what your teacher asks about in class.

1. **Read Part 1 first.** Read each textbook page, then its notes. Don't skip the notes on the reading checks and review questions, because they give you model answers.
2. **Then read Parts 2–5** (after the textbook pages): the other sources the textbook leaves out, a one-page summary, the vocabulary explained in detail, and full answers to the teacher's study questions.
3. **Test yourself.** Cover the answers in Part 5 and try each study question out loud before you check.

Notes are color-coded:

1 Key idea **2 Vocabulary** **3 Make it make sense** **4 Watch out** **5 Connection**

What this guide covers

Where to read

Source	Section	Pages	What it covers
Class textbook: <i>Miller & Levine Biology</i> (2019)	Lesson 1.1 What Is Science?	8–14	observation, inference, hypothesis, data (quantitative and qualitative)
Class textbook: <i>Miller & Levine Biology</i> (2019)	Lesson 1.2 Science in Context	15–18	scientific attitudes, collaboration, feedback, peer review
Reference: <i>Campbell Biology</i>	Ch. 1, Concept 1.3	16–17	observation, data, inductive and deductive reasoning
Teacher's slides	Biology Class Basics (KJ_PH)	slides 3–8	class policies, formative and summative assessment, academic skills
Teacher's slides	Deductive vs Inductive Reasoning Overview	slides 3–6	induction and deduction, with practice examples

Source	Section	Pages	What it covers
Teacher's worksheets	Observation and Inference; Geotastic Inference Game	—	Sherlock Holmes activity, types of data, a prediction-testing game

The unit guide lists no textbook lesson for 1A (1B onward list one). The lessons above were chosen because they cover 1A's vocabulary and study questions. The textbook does not cover class policies or induction and deduction. Those answers use the teacher's slides and Campbell.

What Is Science?

KEY QUESTIONS

- What are the goals of science?
- What procedures are at the core of scientific methodology?
- What is a scientific theory?

VOCABULARY

- 1 observation
inference
hypothesis
controlled experiment
independent variable
dependent variable
control group
data
theory

READING TOOL

List in order the parts of a typical experiment that uses scientific methodology. Use the headings in your text as a guide to fill in the chart in your **Biology Foundations Workbook**.



Humans have wondered about their place in the cosmos since the earliest time. How do we fit into the grand scheme of nature? How much power do we hold over nature? What are the limits to our abilities? These and other questions are more relevant today than ever before. This chapter begins our effort to show you how science tries to answer those questions.

What Science Is and Is Not

This book will help you understand how biologists try to make sense of nature. These pages are filled with many scientific “facts” and ideas. But one of the first things you should understand is that **scientific knowledge is always changing**. Some “facts” and ideas you’ll find here may have changed since this text was written, and others will change soon. Why? Because scientists are constantly testing, debating, and revising scientific explanations of events in the natural world. That constant testing and revising helps explain why scientists **don’t “believe” in scientific facts or ideas. Scientists either understand and accept a particular scientific explanation of the natural world, or they reject that explanation.** But if science is not a list of unchanging facts and beliefs, what is it?

- 4 **The Nature of Science** The term *science* is usually defined as the use of evidence to construct testable explanations and predictions of natural phenomena, as well as the knowledge generated through this process. Most importantly, **science is a process—an organized way of observing and asking questions about the natural world, developing those questions into testable explanations, and gathering and analyzing data that support or reject those explanations.** The word *science* can also refer to the constantly growing and changing body of knowledge that the process of science generates.

Lesson 1.1 What Is Science?

Miller & Levine Biology, textbook page 8

This lesson answers “what is science?” Most of 1A comes from here: observation, inference, and data. The list of words at the top left is the lesson’s vocabulary.

1 LESSON VOCABULARY

For 1A, focus on **observation**, **inference**, and **data**. Hypothesis, controlled experiment, variables, and control group come back in 1B.

2 KEY IDEA

Scientific knowledge changes when new evidence appears. That is why science is a skill you practice, not a list of facts to memorize.

3 WATCH OUT

Scientists don’t “believe” in ideas. They **accept or reject** explanations based on evidence. Avoid writing “scientists believe...” in answers.

4 SCIENCE

Science has two meanings: a **process** (the way we find things out) and the **body of knowledge** that process produces.

5 KEY IDEA

The process in four steps: **observe and ask questions → make a testable explanation → gather and analyze data → support or reject the explanation.** Worth memorizing.

- 6 How is science different from other ways of explaining how the world works? First, science deals only with the natural world. Scientific research never concerns, in any way, supernatural phenomena of any kind. Second, scientists collect and organize information in an orderly way, looking at events for patterns and connections of cause and effect. Third, scientists propose explanations based on evidence and understanding, not belief. Some ways that scientists study the natural world are shown in Figure 1-1.

The Goals of Science Science is based on the view that the physical universe is composed of interacting parts and processes. From a scientific perspective, all objects in the universe, and all interactions among those objects, are governed by **universal natural laws**. The same natural laws apply whether the objects and events are large (like a hurricane) or small (like the cells in your body).

- 8 Greek philosophers were among the first to try to explain the natural world in terms of events and processes they could observe. Modern scientists continue that tradition. **One goal of science is to provide natural and testable explanations for events in the natural world. Science also aims to use explanations supported by data to understand patterns in nature, and to make useful predictions about natural events.**

VIDEO

Discover how scientists use scientific processes to discover the wide diversity of insects that live in people's homes.

INTERACTIVITY CASE STUDY

Figure 1-1 Studying the Natural World

- 9 How do chimpanzees interact with one another? What is the ideal temperature for basil plants in a hydroponics system? What kinds of fish live in the Colorado River? Researchers can use science to answer these questions and many others.



Lesson 1.1 What Is Science?

Miller & Levine Biology, textbook page 9

How science is different from other ways of explaining the world, and what science is trying to do.

6 KEY IDEA

Three features of science: (1) it studies only the **natural** world; (2) it collects information in an **orderly** way and looks for patterns and cause and effect; (3) its explanations come from **evidence**, not belief.

7 MAKE IT MAKE SENSE

The same natural laws apply to something huge (a hurricane) and something tiny (a cell). That's why one experiment can teach us about many situations.

8 GOALS OF SCIENCE

Science aims to (1) give **natural, testable explanations**, (2) understand **patterns** in nature, and (3) make useful **predictions**.

9 CONNECTION

Each question can be answered with data. "Ideal temperature for basil" gives **quantitative** data (numbers). "How chimps interact" gives mostly **qualitative** data (descriptions of behavior).

Science, Change, and Uncertainty Scientists have gathered lots of important information that helps cure and prevent disease, grow food, and link the world electronically. Yet much of nature remains a mystery. Almost every scientific discovery raises more questions than it answers. Often, research yields surprises that point future studies in new and unexpected directions. This constant change doesn't mean science has failed. On the contrary, it shows that science continues to advance.

That's why studying science means more than just memorizing what we know. It also means understanding what we don't know. You may be surprised to hear this, but science rarely "proves" anything. Scientists aim for the best understanding of the natural world that current methods can reveal. Uncertainty is always part of the scientific process, and is part of what makes science exciting!

We hope to show you that understanding science isn't just about learning "facts." We hope you'll gain some understanding of the spirit of scientific inquiry, of the way scientists think, and of both the process and excitement of discovery. Don't just memorize today's scientific facts and ideas. And please don't believe them, just because they are in a textbook! Instead, try to understand how scientists developed those ideas. Pose the kinds of questions scientists ask. Try to see the thinking behind the experiments we describe.

15  **READING CHECK** **Construct an Explanation** How is scientific knowledge different from other types of knowledge?

Scientific Methodology

Is science a mysterious process that only scientists do under special circumstances? Nope! You use scientific thinking all the time!

12 Suppose your family's car won't start. What do you do? You use what you know about cars to ask questions. Is the battery dead? You test that idea by turning the key in the ignition. If the starter motor works but the engine doesn't start, you reject the dead battery idea. Is the car out of gas? A glance at the fuel gauge tests that idea. Again and again, you apply scientific thinking until the problem is solved—or until you run out of ideas and call a mechanic!

13 Scientists work in pretty much the same way. There isn't a single, cut-and-dried "scientific method." But there is a general style of investigation we call scientific methodology, which is a fancy way of saying "the way science works."

14  **Scientific methodology involves observing and asking questions, forming hypotheses, conducting controlled experiments, collecting and analyzing data, and drawing conclusions.** Figure 1-2 shows how one research team used scientific methodology in its study of one particular species of marsh grass in a New England salt marsh.



INTERACTIVITY

Discover the power of scientific methodology.

Lesson 1.1 What Is Science?

Miller & Levine Biology, textbook page 10

Science is uncertain and always improving. The "scientific method" is really a flexible way of thinking that you already use.

10 KEY IDEA

Science doesn't **prove** things. It finds the best explanation the evidence supports. This connects to study question 4: science relies on induction, which is never 100% certain.

11 MAKE IT MAKE SENSE

The book tells you to be **skeptical** of the book! Ask *how* scientists know something, not just *what* they know.

12 EVERYDAY SCIENCE

Each guess ("dead battery?", "out of gas?") is a **hypothesis**, and each check (turn the key, look at the gauge) is a **test**. When a test fails, you reject that idea and try the next one.

13 WATCH OUT

There is **no single fixed** "scientific method" with the same steps every time. Scientists follow a general style called *scientific methodology*.

14 KEY IDEA

Five parts: **observe and ask questions** → **form hypotheses** → **do controlled experiments** → **collect and analyze data** → **draw conclusions**. Figure 1-2 on the next page walks through all five.

15 READING CHECK: ANSWER

Scientific knowledge is based on **evidence** from observation and experiment, it must be **testable**, and it is always open to **revision** when new evidence appears. Other kinds of knowledge (opinion, belief, tradition) don't have to meet those rules.

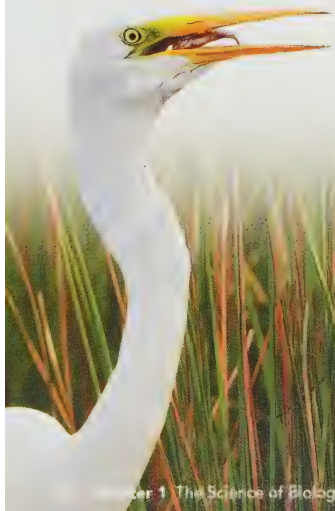
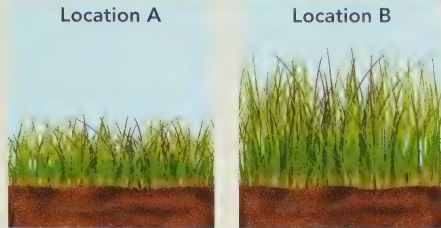


Figure 1-2

Salt Marsh Experiment

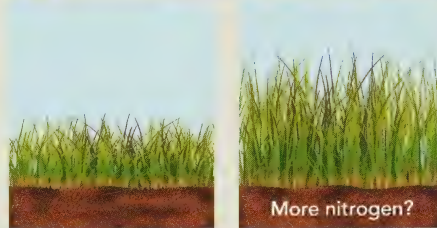
Salt marshes are coastal environments often found where rivers meet the sea. Researchers made an interesting observation about the way a particular species of marsh grass grows. They then applied scientific methodology to answer questions that arose from their observation.

16 1. OBSERVING AND ASKING QUESTIONS



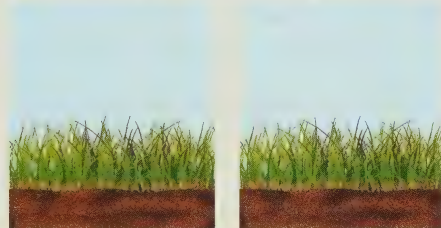
Researchers observed that marsh grass grows taller in some places than others. This observation led to a question: *Why do marsh grasses grow to different heights in different places?*

17 2. INFERRING AND HYPOTHESIZING

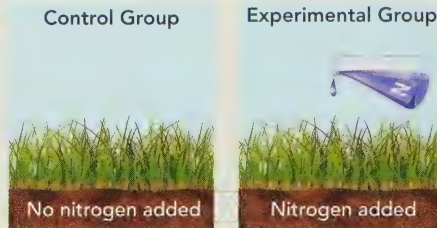


The researchers inferred that something limits grass growth in some places. It could be any environmental factor—temperature, sunlight, water, or nutrients. Based on their knowledge of salt marshes, they proposed a hypothesis: *Marsh grass growth is limited by available nitrogen.*

18 3. DESIGNING CONTROLLED EXPERIMENTS

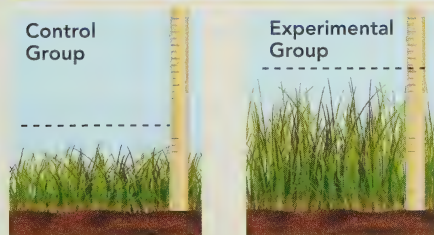


The researchers selected similar plots of marsh grass. All plots had similar plant density, soil type, input of freshwater, and height above average tide level. The plots were divided into control and experimental groups.



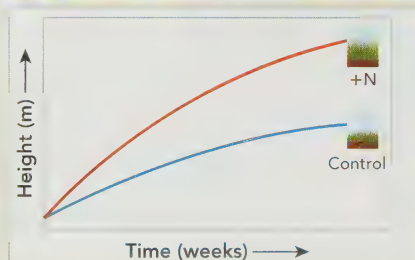
The researchers added nitrogen fertilizer (the independent variable) to the experimental plots. They then observed the growth of marsh grass (the dependent variable) in both experimental and control plots.

19 4. COLLECTING DATA



The researchers sampled all the plots throughout the growing season. They measured growth rates and plant sizes, and analyzed the chemical composition of living leaves.

20 5. ANALYZING CONCLUSIONS



Data from all plots were compared and evaluated by statistical tests. Data analysis confirmed that marsh grasses in experimental plots with additional nitrogen did, in fact, grow taller and larger than controls. The hypothesis and its predictions were supported.

Lesson 1.1 What Is Science?

Miller & Levine Biology, textbook page 11

Figure 1-2 follows one real study from start to finish. Match each numbered step to the 1A vocabulary.

16 STEP 1: OBSERVATION

The **observation** is that grass is taller in some places. It is something you can see and describe. The question it leads to: *why?*

17 STEP 2: INFERENCE

“Something limits growth” is an **inference**, an interpretation of the observation using background knowledge. Narrowing it to one testable cause (nitrogen) makes it a **hypothesis**.

18 CONNECTION

The plots are alike in everything except nitrogen. Only one thing differs between groups: that’s a **controlled experiment** (1B vocabulary).

19 STEP 4: DATA

Growth rates, plant sizes, and leaf chemistry are all **quantitative** data (measured numbers). A note like “the grass was lying flat” would be qualitative.

20 WATCH OUT

Look at the wording: the hypothesis was **supported**, not *proved*. The graph shows the +N plots (red) growing taller than the control (blue) over time.

READING TOOL

Make a chart that lists and describes the different steps of scientific methodology.

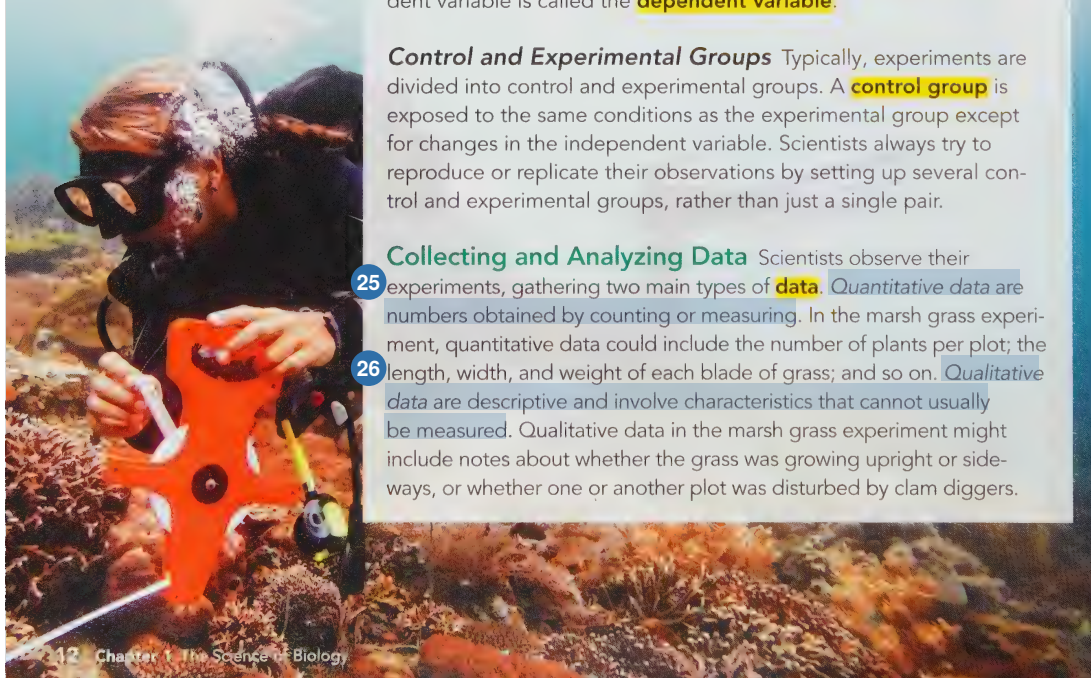


INTERACTIVITY

Try your hand at a simulation of a scientific investigation.

Figure 1-3
Collecting Data

This scientist is collecting data on a coral reef in Indonesia.



- 21 Observing and Asking Questions** Scientific investigations begin with **observation**, the act of noticing and describing events or processes in a careful, orderly way. Of course, scientific observation involves more than just “looking.” The right kind of observation leads to asking questions that no one has asked (or answered) before.

- 23 Inferring and Forming a Hypothesis** After posing questions, scientists use further observations to make inferences. An **inference** is a logical interpretation based on what scientists already know. Inference, combined with a creative imagination, can lead to a hypothesis. A **hypothesis** is a tentative scientific explanation that can be tested by further observation or by experimentation. Scientific hypotheses must then be tested by gathering data that can either support or reject them.

Designing Controlled Experiments Testing hypotheses often involves designing experiments that measure factors that can change. These changeable factors are called variables. Some possible variables include temperature, light, time, and availability of nutrients. Ideally, a hypothesis should be tested by an experiment in which only one variable is changed. All other variables should be kept unchanged, or controlled. This is called a **controlled experiment**.

Controlling Variables It is important to control variables because if several variables are changed, researchers can’t easily tell which variable is responsible for any results they observe. The variable deliberately changed is called the **independent variable**. The variable that is observed and that changes in response to the independent variable is called the **dependent variable**.

Control and Experimental Groups Typically, experiments are divided into control and experimental groups. A **control group** is exposed to the same conditions as the experimental group except for changes in the independent variable. Scientists always try to reproduce or replicate their observations by setting up several control and experimental groups, rather than just a single pair.

- 25 Collecting and Analyzing Data** Scientists observe their experiments, gathering two main types of **data**. **Quantitative data** are numbers obtained by counting or measuring. In the marsh grass experiment, quantitative data could include the number of plants per plot; the length, width, and weight of each blade of grass; and so on. **Qualitative data** are descriptive and involve characteristics that cannot usually be measured. Qualitative data in the marsh grass experiment might include notes about whether the grass was growing upright or sideways, or whether one or another plot was disturbed by clam diggers.

Lesson 1.1 What Is Science?

Miller & Levine Biology, textbook page 12

The most important page for 1A. It defines observation, inference, and the two kinds of data.

21 OBSERVATION

An **observation** is noticing and **describing** carefully, using your senses or instruments. It says *what is there*, not what it means.

22 MAKE IT MAKE SENSE

Good observations lead to good **questions**. Sherlock Holmes is famous because he notices details other people miss.

23 INFERENCE

An **inference** goes *beyond* the observation to explain it, using what you already know. It can be wrong even when the observation is right.

24 KEY IDEA

The chain: **observation** → **inference** → **hypothesis** → **test**. A hypothesis is an inference stated so it can be tested.

25 QUANTITATIVE DATA

Numbers from counting or measuring. *Quanti*tative = *quanti*ty. Examples: 47 plants, 30 cm tall, 22 °C.

26 QUALITATIVE DATA

Descriptions instead of numbers. *Quali*tative = *quali*ty (what kind). Examples: upright or sideways grass, yellow leaves, a plot disturbed by clam diggers.

Selecting Equipment and Technology Scientists collect and analyze data using tools that range from simple meter sticks to complex hardware that measures leaf nitrogen content. Data are often gathered directly by hardware controlled by computers running software that organizes and analyzes results. **Statistical analysis helps determine if an experimental treatment is significantly different from controls.**

Sources of Error Researchers must avoid errors in data collection and analysis. Tools used to measure the size and mass of marsh grasses, for example, have limited accuracy. Data analysis and sample size must be chosen carefully. **The larger the sample size, the more reliably researchers can analyze variation within each group, and evaluate differences between experimentals and controls.**

29 Interpreting Data and Drawing Conclusions Data analysis may lead to conclusions that support or refute the hypothesis being tested. Often, new data indicate that a hypothesis is on the right track, but is off-base about a few details. New questions lead to new and revised hypotheses, which are tested with new experiments that involve better control of variables or other changes in experimental design.

30 When Experiments Aren't Possible Not all hypotheses can be tested by experiments. Animal behavior researchers, for example, might propose hypotheses about how groups of animals interact in nature. These hypotheses are tested by field observations designed to disturb natural behavior as little as possible. Analysis of data from these observations may lead to new hypotheses that can be tested in different ways. If investigations suggest, for example, that members of a group are related to one another, genetic tests can gather data that support or reject that hypothesis.

Sometimes ethics prevent certain types of experiments. For example, in medical research, when a chemical is suspected as a cause of cancer, researchers do not purposely expose volunteers to the chemical. They study people who have already been exposed, using those who have not as the control group.

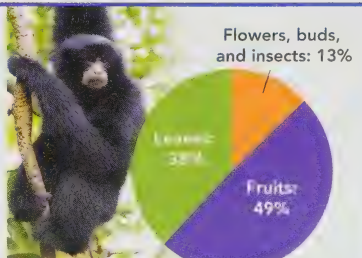
31 READING CHECK Describe What is the difference between quantitative data and qualitative data?

Analyzing Data

What's in a Diet?

The circle graph shows the diet of the siamang gibbon, a type of ape found in the rain forests of Southeast Asia.

- 32 1. Analyze Graphs** According to the circle graph, which plant parts do siamangs rely on most as a food source?
- 2. Form a Hypothesis** How would siamangs be affected if the rain forests they live in produced less fruit? Explain your reasoning.



Lesson 1.1 What Is Science?

Miller & Levine Biology, textbook page 13

How scientists make their data trustworthy, and what to do when an experiment isn't possible.

27 MAKE IT MAKE SENSE

Statistics check whether a difference is real or just chance. If the nitrogen plots were only 1 cm taller, that might be luck.

28 KEY IDEA

More data make conclusions more reliable. That's the idea behind the Geotastic game: more advances (more observations) should give better location guesses (better inferences).

29 KEY IDEA

Conclusions **support or refute** (reject) a hypothesis. Often the result is "partly right," which leads to a revised hypothesis and a better experiment. Science improves the same way you do: test, get feedback, revise.

30 MAKE IT MAKE SENSE

Some questions (like animal behavior in the wild) are answered by careful **field observations** instead of experiments. Jane Goodall's chimpanzee research is a famous example.

31 READING CHECK: ANSWER

Quantitative data are numbers from counting or measuring (height = 30 cm). **Qualitative** data are descriptions of characteristics that usually can't be measured (the grass is leaning sideways).

32 ANALYZING DATA: ANSWERS

1. **Fruits** (49% of the diet; leaves are 38%). 2. With less fruit, siamangs would have less of their main food. They might eat more leaves, lose weight, or have fewer young. A hypothesis: *if fruit decreases, then siamang populations will decline.* The graph is **quantitative** data.

Scientific Theories

BUILD VOCABULARY

ACADEMIC WORDS A

scientific **theory** describes a well-tested explanation for a range of phenomena. Scientific theories are different from scientific laws, and it is important to understand that theories do not become laws. Laws, such as ideal gas laws in chemistry or Newton's laws of motion, are concise, specific descriptions of how some aspects of the natural world are expected to behave in a certain situation. In contrast, scientific theories, such as cell theory or the theory of evolution, are more dynamic and complex. Scientific theories encompass a greater number of ideas and hypotheses than laws, and are constantly fine-tuned through the process of science.

Researchers hope that data from many individual experiments add up to something bigger—a larger and more useful understanding of how the world works. That's where we encounter an important scientific term: **theory**. Many other terms you'll learn this year will be new to you, because they are used only in science. But the word *theory* is also often used in everyday life. That causes problems, because non-scientists use the word *theory* in a very different way than scientists do. **33** When most people say, "I have a theory," they mean, "I have a hunch" or "I have a guess." When a friend says, "That's just a theory," she may mean, "People aren't too certain about that idea." Scientists would never use the word *theory* in that way. In those situations, a scientist would use the common words "guess" or "hunch" or a scientific term we've already discussed: *hypothesis*.

When scientists refer to gravitational theory or evolutionary theory, they do not mean "a hunch about gravity" or "a guess about evolution." **34** *In science, the word theory applies to a tested, highly-reliable scientific explanation of events in the natural world that unifies many repeated observations and incorporates durable, well-supported hypotheses that enable scientists to make accurate predictions.* Charles Darwin developed lots of hypotheses over many years. It took a long time for him to assemble his thoughts and hypotheses into his theory of evolution by natural selection. Since then, evolutionary theory has predicted things Darwin couldn't have imagined, such as the evolution of bacteria that resist antibiotics and insects that are immune to pesticides. Today, evolutionary theory is the central organizing principle of all biological science.

Once a theory has been thoroughly tested and supported by many lines of evidence, it may become the dominant scientific view. **35** But remember that no theory is absolute truth. Science is always changing; as new evidence is uncovered, a theory may be revised or replaced by a more useful explanation.

LESSON 1.1 Review

KEY IDEAS

- 36** 1. In your own words, define the term *science*.
2. Why are hypotheses so important to controlled experiments?
3. How does a theory differ from a hypothesis?

CRITICAL THINKING

- 37** 4. **Form a Hypothesis** You observe mold growing on one side of a slice of bread, but not on the other side. Form a hypothesis to explain this difference in mold growth.

5. **Plan an Investigation** Design a controlled experiment to test the effect of water temperature on goldfish. Be sure to include your hypothesis, independent variable, and dependent variable, as well as the experimental and control groups in your experiment.

Lesson 1.1 What Is Science?

Miller & Levine Biology, textbook page 14

What a scientific theory is, plus the lesson review. The notes give model answers to every review question.

33 WATCH OUT

In everyday speech, "theory" means a guess. In science it means the **opposite**: a well-tested, highly reliable explanation.

34 THEORY

A **theory** is a tested, highly reliable explanation that ties together many observations and well-supported hypotheses, and that makes accurate predictions. Examples: cell theory, evolution.

35 KEY IDEA

Even a strong theory can be **revised** if new evidence appears. Again: science supports, it doesn't prove.

36 REVIEW 1–3: ANSWERS

1. Science is an organized way of using evidence to explain and predict events in the natural world, plus the knowledge that process builds. 2. A hypothesis gives the experiment its purpose: it says what to change (independent variable) and what result to expect. 3. A hypothesis is a single testable explanation. A theory is a well-tested explanation that unifies many observations and hypotheses.

37 REVIEW 4–5: ANSWERS

4. Example: *the moldy side got more moisture* (or less light, or touched a mold source). 5. Hypothesis: warmer water increases goldfish activity. Independent variable: water temperature (e.g., 15, 20, 25 °C). Dependent variable: gill beats per minute. Control group: fish at normal tank temperature. Keep fish size, tank size, and food the same, and use several fish per group.

Science in Context

LESSON 1.2



KEY QUESTIONS

- What attitudes and experiences generate new ideas?
- Why is peer review important?
- What is the relationship between science and society?
- What practices are shared by science and engineering?

VOCABULARY

40 **bias**

READING TOOL

Before you read, study **Figure 1-4**. As you read, list examples of each of the different aspects of science in the table in your **Biology Foundations Notebook**. Then use **Figure 1-9** to add examples for engineering.

38 The testing of ideas is the heart of science and engineering. In their quest for understanding, scientists engage in many different activities: They ask questions, make observations, seek evidence, share ideas, and analyze data. These activities are all part of the dynamic process of science.

39 The process of science typically consists of the components shown in **Figure 1-4**. Notice that the parts of the process do not proceed in a linear fashion. Real science usually involves many activities that loop back on themselves, building up knowledge as they proceed. In fact, science is at its heart a creative endeavor. Scientists take many different paths through the process depending on the questions they are investigating and the resources available to them.



Adapted from *Understanding Science*, UC Berkeley, Museum of Paleontology

INTERACTIVITY

Figure 1-4 The Process of Science

As the arrows indicate, the different aspects of science are interconnected—making the process of science dynamic, flexible, and unpredictable.

Lesson 1.2 Science in Context

Miller & Levine Biology, textbook page 15

Lesson 1.2 shows science as a messy loop, not a straight line, and as something done by a community.

38 KEY IDEA

Scientists question, observe, look for evidence, **share ideas**, and analyze data. All of these are part of science, not just experiments.

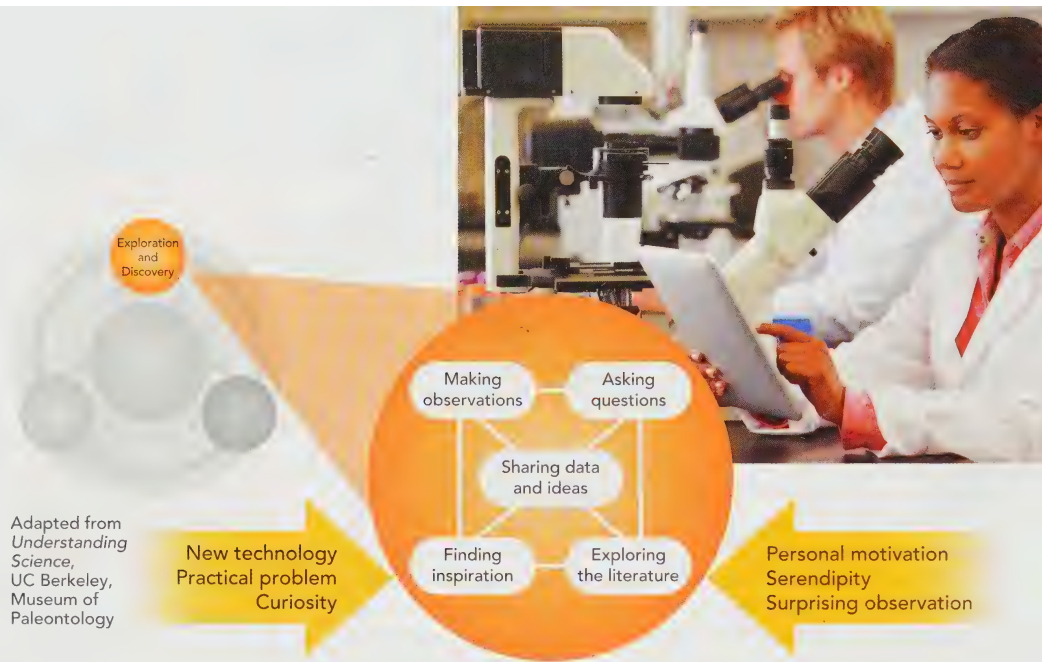
39 KEY IDEA

Real science **loops back** on itself. Results send you back to new questions. This is the same loop as deliberate practice in 1A: try, get feedback, fix, try again.

40 BIAS

A **bias** is “a personal, rather than scientific, point of view for, or against, something” (p. 20). Controlled experiments and peer review help keep bias out of results.

Where scientific ideas come from, and the habits of mind that good scientists (and good students) use.



Adapted from
Understanding Science,
UC Berkeley,
Museum of
Paleontology

Figure 1-5
**Exploration and
Discovery**

41 Ideas in science can arise in many ways—from simple curiosity or from the need to solve a particular problem. Scientists often begin investigating by making observations, asking questions, talking with colleagues, and reading about previous experiments.

READING TOOL

Copy one of the concept maps in this lesson. Then, add specific examples to illustrate the processes it describes.

Exploration and Discovery

Usually, the testing of ideas begins with observations and questions. Those observations and questions come from a variety of different sources, as shown in Figure 1-5.

Scientific Attitudes Good scientists share scientific attitudes, or habits of mind, that lead them to exploration and discovery.

42 *Curiosity, skepticism, open-mindedness, and creativity help scientists and engineers ask new questions and define new problems.*

Curiosity A scientist may look at a salt marsh and ask, “What’s that plant? Why is it growing here?” Previous studies can spark curiosity and lead to new questions. Engineering problems often arise from practical issues involving humans and health or environmental issues.

43 **Skepticism** Scientists and engineers should be skeptics who question existing ideas and hypotheses, and refuse to accept explanations without evidence. Scientists often design new experiments to test hypotheses proposed by other scientists. Engineers work to solve problems in the physical world using scientific thinking.

44 **Open-Mindedness** Scientists and engineers should be open-minded, meaning that they are willing to accept different ideas that may not agree with their hypotheses.

Creativity Creativity and critical thinking are essential for asking questions, defining problems, proposing hypotheses, and designing experiments that yield accurate data.

41 CONNECTION

Scientists start by observing, asking questions, **talking with colleagues**, and reading earlier work. That is **collaboration** (a 1A academic word).

42 KEY IDEA

The four **scientific attitudes**. Know all four, and be ready to give an example of each.

43 SKEPTICISM

Being **skeptical** means questioning ideas and asking for evidence before accepting them. It’s not the same as refusing to believe anything.

44 OPEN-MINDEDNESS

Being **open-minded** means being willing to accept evidence and ideas that disagree with what you expected. Pairs well with skepticism: question everything, but change your mind when the evidence says so.

Practical Problems Ideas for scientific investigations often arise from practical problems involving humans and health or environmental issues. Salt marshes, for example, are vital nurseries for commercially important fish and shellfish. Yet salt marshes are under intense pressure from agriculture and from development. An engineer might wonder, “How would nearby construction affect the marsh and how can that impact be minimized?”

New Technology New technology often opens new ways of asking questions for both scientists and engineers. For example, portable, remote data-collecting equipment enables field researchers to monitor environmental conditions around the clock, in several different locations at once. This information allows researchers to pose and test new hypotheses.



VIDEO

Listen to scientists discussing their work in extreme environments.

- 48 **READING CHECK** **Describe** Describe a “fact” that you have heard or read that made you skeptical. Explain your reasons for doubt.

Community Analysis and Feedback

- 45 Scientists often collaborate in groups and communicate with other research groups. In order for research to be accepted, however, it must be officially shared with the scientific community. The communication must follow a variety of rules that ensure it is scientifically appropriate. Some of the steps that scientists use when communicating their results are shown in **Figure 1-6**.

- 47 Scientists also share their work with the general public, especially when the results or information could benefit society. This communication is less formal, and can take the form of a newspaper or magazine article, television program, or blog post.

Figure 1-6

Community Analysis and Feedback

- 46 Communication is an important part of science. Scientists review and evaluate one another’s work to ensure accuracy. Results from one study may lead to new ideas and further studies.

Lesson 1.2 Science in Context

Miller & Levine Biology, textbook page 17

How scientists work together and check each other’s work.

45 COLLABORATION

Scientists **collaborate**: they work in teams and share with other teams. Results only count once they are shared with the scientific community.

46 FEEDBACK

Figure 1-6: scientists give each other **feedback** through peer review, discussion, and replication. Your formative assessments do the same job for you.

47 MAKE IT MAKE SENSE

Scientists also share informally, through news articles, TV, and blogs, especially when results could help society.

48 READING CHECK: ANSWER

Example: “Cracking your knuckles causes arthritis.” A skeptic asks what evidence supports it. (Studies comparing knuckle-crackers with non-crackers have not found more arthritis.)



Adapted from *Understanding Science*, UC Berkeley, Museum of Paleontology



53 Replicating Procedures

1. Working with a partner, assemble ten blocks into an unusual structure. Write directions that others can use to replicate that structure without seeing it.
2. Exchange directions with another team. Replicate the team's structure by following their directions.
3. Compare each replicated structure to the original. Identify which parts of the directions were clear and accurate, and which were unclear or misleading.

ANALYZE AND CONCLUDE

1. **Evaluate** How could you have written better directions?

2. **Construct an Explanation** Based on what you have learned in this investigation, explain why it is important that scientists write procedures that can be replicated.



Figure 1-7
Mangrove Swamp

In tropical areas, mangrove swamps serve as the ecological equivalents of temperate salt marshes. The results of the salt marsh experiment suggest that nitrogen might be a limiting nutrient for mangroves and other plants in these similar habitats.

- 49 **Peer Review** Scientists share findings with other scientists by publishing their hypotheses, experimental methods, results, and analysis in scientific journals. Papers submitted to these journals are reviewed by anonymous, independent reviewers who work in the same field. Reviewers look carefully for mistakes, oversights, unfair influences, or fraud, in techniques and analysis. Their goal is to ensure that articles meet the highest standards of quality. This process is called peer review. *Publishing articles in peer-reviewed scientific journals allows researchers to share ideas. It also allows other scientists to evaluate and test the data and analysis.* Peer review does not guarantee that a piece of work is correct, but it does certify that the work meets standards set by the scientific community.

Sharing Knowledge and New Ideas After research has been published, it enters the dynamic marketplace of scientific ideas. How do new data fit into existing scientific understanding? For example, the finding that growth of salt marsh grasses is limited by available nitrogen sparks more questions: Is the growth of other plants in the same habitat also limited by nitrogen? What about the growth of different plants in similar environments, such as the mangrove swamp shown in **Figure 1-7**? These logical and important questions leads to new hypotheses that must be independently supported or rejected by controlled experiments.

- 51
- 52 **READING CHECK Explain** Why is it necessary for other scientists to evaluate the findings of scientists in their field?

Lesson 1.2 Science in Context

Miller & Levine Biology, textbook page 18

Peer review: how the scientific community checks new research before accepting it.

49 PEER REVIEW

Peer review: other experts in the same field check a paper for mistakes, bias, and fraud before it is published. It's feedback and assessment for scientists.

50 WATCH OUT

Peer review doesn't **guarantee** a paper is right. It shows the work meets the community's standards.

51 KEY IDEA

Every answer raises **new questions** (does nitrogen limit mangroves too?). Science keeps going.

52 READING CHECK: ANSWER

Other experts can catch mistakes, bias, or fraud the authors missed, and can **replicate** the work to see if they get the same result. That makes the findings more trustworthy.

53 CONNECTION

Quick Lab: if your directions are unclear, others can't **replicate** your work. This is the 1A academic skill of writing clearly and concisely.

Part 1 (continued). Other sources for this target

The class textbook pages are reproduced in full above. These excerpts cover the rest: the reference book *Campbell Biology* and anything the class textbook leaves out.

Campbell Biology, Chapter 1, Concept 1.3

The teacher's *Observation and Inference* worksheet asks questions about this passage ("The scientific process begins with?", "Recorded observations are called?").

Observation (p. 16)

Biology, like other sciences, begins with careful observations. In gathering information, biologists often use tools such as microscopes, precision thermometers, or high-speed cameras that extend their senses or facilitate careful measurement. ... In exploring nature, biologists also rely heavily on the scientific literature, the published contributions of fellow scientists.

Data (p. 16)

Recorded observations are called data. Put another way, data are items of information on which scientific inquiry is based. The term *data* implies numbers to many people. But some data are qualitative, often in the form of recorded descriptions rather than numerical measurements. For example, Jane Goodall spent decades recording her observations of chimpanzee behavior ... Quantitative data are generally expressed as numerical measurements and often organized into tables and graphs.

Inductive reasoning (p. 17)

Collecting and analyzing observations can lead to important conclusions based on a type of logic called inductive reasoning. **Through induction, we derive generalizations from a large number of specific observations.** "The sun always rises in the east" is one example. Another biological example is the generalization "All organisms are made of cells," which was based on two centuries of microscopic observations ...

Deductive reasoning (p. 17)

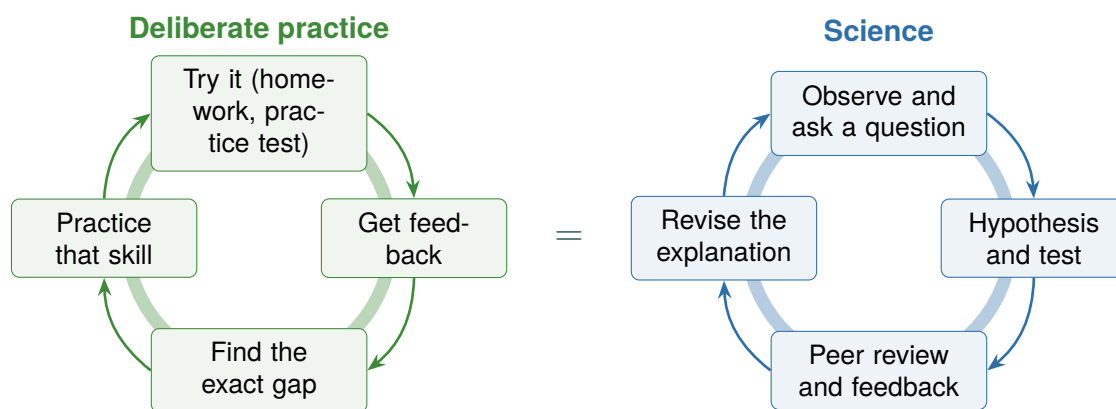
While induction entails reasoning from a set of specific observations to reach a general conclusion, **deductive reasoning involves logic that flows in the opposite direction, from the general to the specific.** ... In the scientific process, deductions usually take the form of a prediction of results that will be found if a particular hypothesis (premise) is correct. ... This deductive testing takes the form of "If . . . then" logic. In the case of the desk lamp example: If the burnt-out bulb hypothesis is correct, then the lamp should work if you replace the bulb with a new one.

A hypothesis is never proved (p. 17)

Although a hypothesis can never be proved beyond all doubt, testing it in various ways can significantly increase our confidence in its validity.

Part 2. Summary

The big idea of 1A. Science is a *skill*, not a list of facts, and skills improve through deliberate practice. Deliberate practice means: try it, get feedback, fix the gap, try again. That is the same loop scientists use, where they observe, explain, test, get feedback from peers, and revise. This first target teaches two sets of skills together:



Deliberate practice (left) and science (right) run on the same loop: try, get feedback, fix, try again.

1. How science thinks.

- Science begins with **observation**: noticing and describing carefully, using senses and instruments.
- Recorded observations are **data**. **Quantitative** data are numbers (counts, measurements). **Qualitative** data are descriptions (color, behavior, shape).
- From observations, scientists make **inferences**, logical interpretations based on what they already know. An inference plus creativity can become a testable **hypothesis**.
- **Induction** moves from many specific observations to a general conclusion ("every swan I've seen is white → all swans are white"). Its conclusion is *probably* true, never guaranteed.
- **Deduction** moves from a general statement to a specific prediction ("if nitrogen limits growth, then adding nitrogen should make grass taller"). If the premises are true, the conclusion must be true.
- Science uses both. Induction builds the general idea, and deduction turns it into an if-then prediction you can test. Because the general idea came from induction, science never fully "proves" anything. Evidence makes an explanation more or less likely.
- Science is a community effort: **collaboration**, **feedback**, and **peer review** catch mistakes and improve ideas.

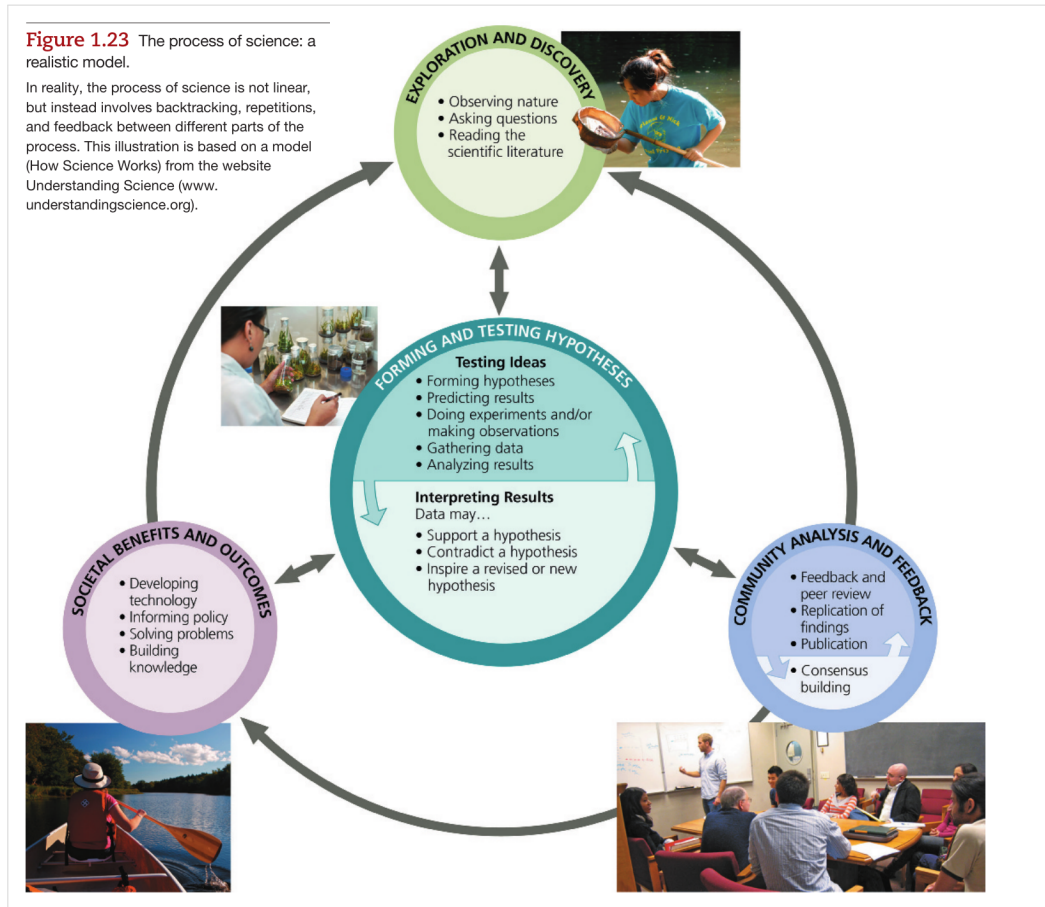
1. How this class works.

- The **KUDOS** and **Unit Guide** tell you what to learn: vocabulary, where to find information, and study questions.
- **Formative** assessments (homework, practice tests, CFAs) are practice. They don't affect your grade and exist to give you feedback.
- **Summative** assessments (quizzes, unit exams, final) measure what you learned and count toward your grade.
- Academic skills you are expected to practice: note taking, reading comprehension, **annotation**, clear and concise writing, academic vocabulary, CER argumentation, and study and test-taking strategies.

One-sentence version: observe carefully, label what you *saw* separately from what you *think it means*, reason from evidence, and use feedback to keep improving, both in science and in this class.

Figure 1.23 The process of science: a realistic model.

In reality, the process of science is not linear, but instead involves backtracking, repetitions, and feedback between different parts of the process. This illustration is based on a model (How Science Works) from the website Understanding Science (www.understandingscience.org).



The process of science is a loop, not a straight line. Testing ideas (center) connects to exploring (top), community feedback (right), and benefits to society (left). Deliberate practice works the same way.

Source: Campbell Biology, Figure 1.23, p. 18

Part 3. Biology vocabulary

annotation

- **Definition:** Notes you write directly on a text, diagram, or data table while reading. You mark key ideas, define unfamiliar words, ask questions, and make connections.
- **Source:** Biology Class Basics, slide 8 (listed as an academic skill). The textbook doesn't define it, though its Reading Tool boxes ask you to annotate.
- **What good annotation looks like:**
 - Underline or highlight the *one* key sentence in a paragraph, not the whole paragraph. (In Miller & Levine, the key sentence often starts with a small key icon.)
 - Write a short note in the margin in your own words: "inference = interpretation, not observation."
 - Put "?" next to anything confusing and write the question down.
 - Draw arrows to connect ideas: "→ same as the marsh grass example."
- **Why it matters:** It turns reading into active thinking, which is deliberate practice. You also remember more, and your notes make review faster before a test.
- **Common mistake:** highlighting everything. If everything is marked, nothing stands out.

After posing questions, scientists use further observations to make inferences.¹

An inference is a logical interpretation based on what scientists already know.² Inference, combined with a creative imagination, can lead to a hypothesis.³ A hypothesis is a tentative scientific explanation that can be tested by further observation or by experimentation.⁴

1 Define: inference = what I *conclude*, not what I see

2 Key sentence: highlight only this one

3 Question: how is a hypothesis different from an inference?

4 Connect: same as the salt-marsh example on p. 11

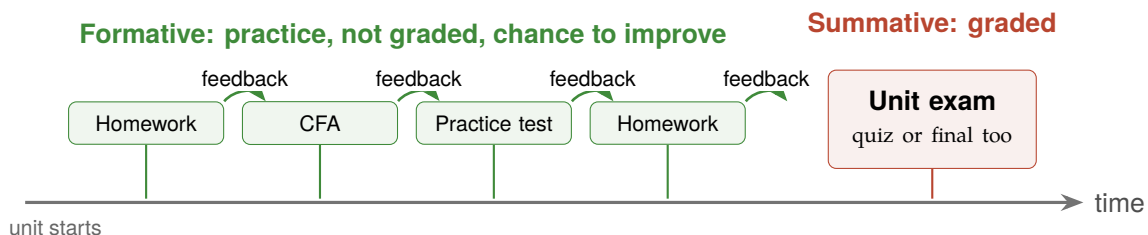
A sample of good annotation: one key sentence marked, a word defined in the margin, a question, and a connection.

formative (formative assessment)

- **Definition:** An assessment used *during* learning to check understanding and give feedback. It is practice, not a final score.
- **In this class:** homework, practice tests, and CFAs (common formative assessments). According to the Class Basics slides, formative assessments **do not affect your grade** and are a “chance for feedback and improvement.”
- **Memory hook:** *form*ative helps *form* your learning, like clay still being shaped.
- **How to use it:** treat each formative as information. Every mistake tells you exactly what to practice before the real test.

summative (summative assessment)

- **Definition:** An assessment at the *end* of a period of learning that measures what you learned and counts toward your grade.
- **In this class:** quizzes, unit exams, and the final exam. Quizzes/exams are 45% of the grade and the final is 20%.
- **Memory hook:** *sum*ative is the *sum* of what you learned.
- **Formative vs. summative:** the same kind of question can appear on both. What differs is the *purpose*: to practice and get feedback (formative) or to measure and grade (summative).



Across a unit: many formative checks give feedback while you are still learning; the summative exam measures what you learned.

quantitative / qualitative (data)

- **Quantitative data:** “numbers obtained by counting or measuring” (M&L p. 12). Examples: number of plants per plot, blade length in cm, temperature in °C, a heart rate of 72 beats per minute.

- **Qualitative data:** “descriptive and involve characteristics that cannot usually be measured” (M&L p. 12). Examples: grass growing upright or sideways, leaf color is yellow, the chimp groomed another chimp (Jane Goodall’s notes, Campbell p. 16).
- **Memory hook:** *quanti*tative = *quanti*ty (how many, how much); *quali*tative = *quali*ty (what kind, what it’s like).
- **Same object, both kinds:** a microscope can count blood cells (quantitative) or show the steps of cell division (qualitative). The worksheet uses this example.
- **Watch out:** a description with a number-like word is still qualitative if nothing was counted or measured. “Many plants” is qualitative; “47 plants” is quantitative.



Qualitative data: Jane Goodall recorded chimpanzee behavior as written descriptions and sketches.

Source: Campbell Biology, Figure 1.21, p. 16

data

- **Definition:** “Recorded observations are called data” (Campbell p. 16). Data are the information scientific inquiry is based on, collected with senses or instruments.
- **Two types:** quantitative and qualitative (above).
- **Role in science:** data *support or reject* a hypothesis (M&L p. 8). Scientists often organize data in tables and graphs and use statistics to check whether a result is real or just random chance (M&L p. 13; Campbell p. 16).
- **Grammar note:** *data* is plural (the singular is *datum*), so scientists write “the data show ...”
- **Good data are:** recorded carefully, repeated (replicated), and taken from a large enough sample. The larger the sample size, the more reliable the analysis (M&L p. 13).

Figure 1.25

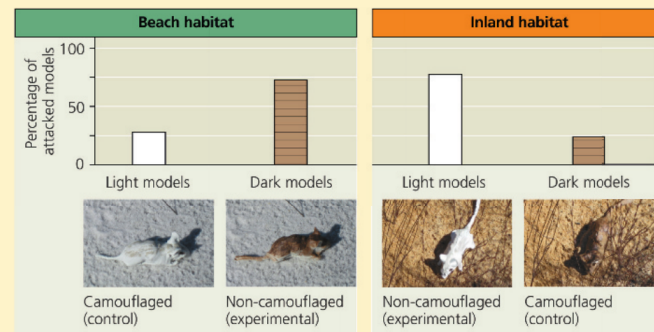
Inquiry: Does camouflage affect predation rates on two populations of mice?

Experiment

Hopi Hoekstra and colleagues tested the hypothesis that coat coloration provides camouflage that protects beach and inland populations of *Peromyscus polionotus* mice from predation in their habitats. The researchers spray-painted mouse models with light or dark color patterns that matched those of the beach and inland mice and placed models with each of the patterns in both habitats. The next morning, they counted damaged or missing models.

Results

For each habitat, the researchers calculated the percentage of attacked models that were camouflaged or non-camouflaged. In both habitats, the models whose pattern did not match their surroundings suffered much higher "predation" than did the camouflaged models.



Data from S. N. Vignieri, J. G. Larson, and H. E. Hoekstra, The selective advantage of crypsis in mice, *Evolution* 64:2153–2158 (2010).

Conclusion

The results are consistent with the researchers' prediction: that mouse models with camouflage coloration would be attacked less often than non-camouflaged mouse models. Thus, the experiment supports the camouflage hypothesis.

INTERPRET THE DATA The bars indicate the percentage of the attacked models that were either light or dark. Assume 100 mouse models were attacked in each habitat. For the beach habitat, how many were light models? Dark models? Answer the same questions for the inland habitat. Do

Quantitative data: the percentage of mouse models attacked by predators, shown as a bar graph. Camouflaged models were attacked much less.

Source: Campbell Biology, Figure 1.25, p. 20

inference

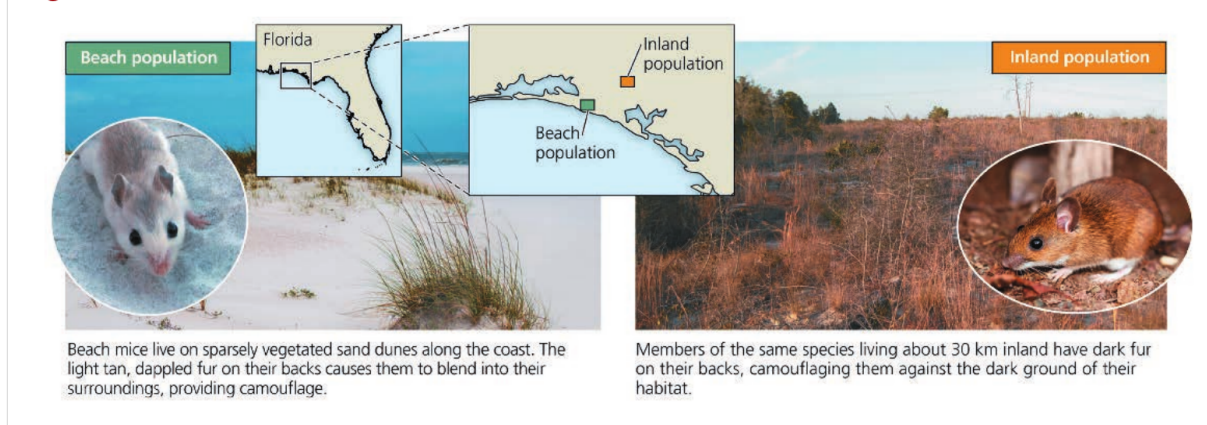
- **Definition:** "a logical interpretation based on what scientists already know" (M&L p. 12). An inference goes *beyond* what was directly observed to explain it.

- **Observation vs. inference:**

| Observation (what you detect) | Inference (what you conclude it means) | |—|—| | The grass in location A is 90 cm tall; in location B it is 30 cm. | Something in location B is limiting growth. | | The road signs are in Portuguese. (Geotastic) | This is probably Brazil or Portugal. | | There is a tan line on the man's wrist, but not his hand. (Sherlock) | He has recently been somewhere sunny. | | The lamp won't light. | The bulb is probably burnt out. |

- **Key point:** an inference can be wrong even when the observation is right. That's why scientists turn inferences into hypotheses and test them.
- **Test tip:** if a statement contains "because," "probably," "must be," or a cause, it's usually an inference.

Figure 1.24 Different coloration in beach and inland populations of *Peromyscus polionotus*.



Observation: beach mice are light and inland mice are dark, each matching its surroundings. Inference: the coloring camouflages them from predators. That inference became a hypothesis, tested in Figure 1.25.

Source: Campbell Biology, Figure 1.24, p. 19

Part 4. Academic vocabulary

These are general academic words, not just biology words. You'll see them on tests and in instructions.

assess / assessment

- **Meaning:** to *assess* is to judge or measure the quality, amount, or level of something. An *assessment* is the tool or event that does the judging: a test, quiz, checkup, or lab report.
- **In class:** formative and summative are the two kinds of assessment.
- **Example sentence:** "The teacher used a checkup to *assess* whether students completed the homework."
- **In science:** scientists assess evidence to decide whether it supports a hypothesis. Peer reviewers assess each other's papers.

collaboration

- **Meaning:** working together with others toward a shared goal. (Verb: *collaborate*; adjective: *collaborative*.)
- **In science:** "Scientists often collaborate in groups and communicate with other research groups" (M&L p. 17). Discussing with colleagues, sharing data, replication, and peer review are all forms of collaboration (M&L Figure 1-6).
- **In class:** lab partners, group discussion, sharing class data in a Google Sheet (Geotastic).
- **Caution:** collaboration means building understanding together, not copying. You still need to be able to explain the idea on your own.



Community analysis and feedback: scientists collaborate through peer review, replication, publication, and discussion.

Source: Campbell Biology, Figure 1.23 (detail), p. 18

commitment

- **Meaning:** a promise or dedication to keep doing something, especially when it's hard. (Verb: *commit*.)
- **In class:** 15–30 minutes of *focused* homework per night (Class Basics slide 7), coming to tutorial when stuck, and doing formative practice even though it isn't graded.
- **In science:** Jane Goodall's decades of observing chimpanzees (Campbell p. 16), and Darwin spending many years assembling his hypotheses into a theory (M&L p. 14).
- **Link to 1A:** deliberate practice only works with commitment, meaning repeated, focused effort over time.

effective

- **Meaning:** successful in producing the intended result. (Noun: *effectiveness*.)
- **Don't confuse:** *effective* (it works) and *efficient* (it works without wasting time or energy). Rereading notes for an hour can be *inefficient* and *ineffective*. Quizzing yourself for 20 minutes is usually more effective.
- **Example sentence:** "Annotating the reading was an *effective* way to prepare for the quiz."
- **In science:** a controlled experiment is an effective design because changing only one variable lets you tell which variable caused the result (M&L p. 12).

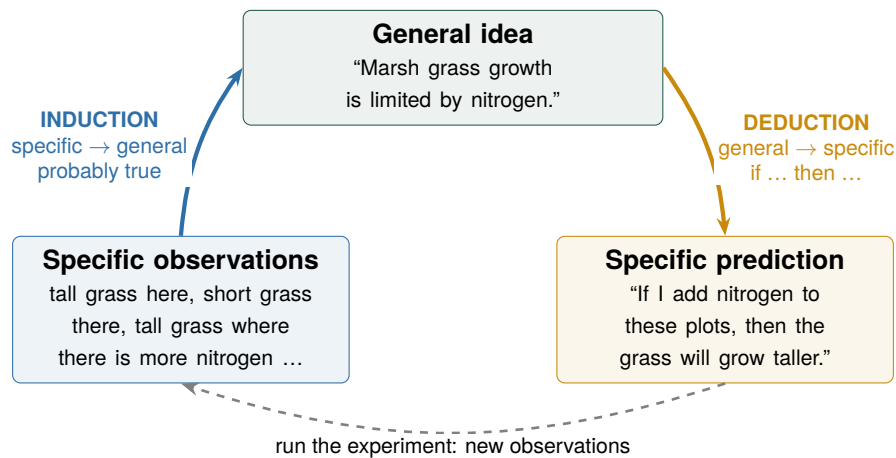
feedback

- **Meaning:** information about your performance that tells you what is working and what to improve.
- **In class:** formative assessments are a "chance for feedback and improvement" (Class Basics slide 6). Comments on assignments and retake opportunities are also feedback.
- **In science:** "Community Analysis and Feedback" (M&L p. 17). Peer reviewers give feedback that catches "mistakes, oversights, unfair influences, or fraud" (M&L p. 18).

- **How to use it:** (1) read the feedback, (2) find the specific gap, (3) practice that exact skill, (4) check again. That loop *is* deliberate practice.
- **Heads-up for 1B:** in biology, *feedback* also means something different: a *feedback loop* in the body (negative and positive feedback) that keeps conditions stable. Same word, related idea: output information is used to adjust the next step.

induction / deduction

- **Induction (inductive reasoning):** reasoning from **specific observations** → **a general conclusion**. Campbell p. 17: “we derive generalizations from a large number of specific observations.”
 - Example: “Every time I drop an apple it falls. → All dropped apples will fall.” (teacher’s slide 5)
 - Example: “Two centuries of microscope observations found cells in every organism → All organisms are made of cells.” (Campbell p. 17)
 - The conclusion is only *probably* true. New evidence could overturn it.
- **Deduction (deductive reasoning):** reasoning from **general premises** → **a specific conclusion**. Campbell p. 17: “from the general to the specific.”
 - Example: “All antibiotics kill bacteria. Penicillin is an antibiotic. Therefore penicillin kills bacteria.” (teacher’s slide 5)
 - If the premises are true and the logic is valid, the conclusion **must** be true. It can only be false if a premise is false.
 - In science, deduction takes the form of an **if ... then** prediction.
- **Memory hook:** Induction = going **into** a big idea from small pieces (specific → general). Deduction = going **down** from the big idea to one case (general → specific).



Induction climbs from specific observations to a general idea; deduction comes back down to a specific prediction you can test. Science cycles between them.

- **Quick check** (teacher’s slide 5):
 - “All dolphins are mammals. All mammals have kidneys. Dolphins have kidneys.” → *deductive*
 - “Chimps and humans share 99% of their genetic material. Chimps and humans evolved from a common ancestor.” → *inductive*
 - “5, 8, 11, 14, 17, __. The next number is 20.” → *inductive* (a pattern is assumed to continue)

observation

- **Meaning:** “the act of noticing and describing events or processes in a careful, orderly way” (M&L p. 12). Also the thing noticed (“my observation was ...”).
- **How we observe:** with the five senses, extended by instruments such as microscopes, thermometers, high-speed cameras, rulers, scales, and pH meters (Campbell p. 16). Scientists also rely on others’ published observations, the scientific literature (Campbell p. 16).
- **Good observations are:** specific, factual, and free of interpretation. “The leaf is yellow with brown edges” is an observation. “The leaf is dying” is an inference.
- **Why it matters:** observation starts every investigation, and “the right kind of observation leads to asking questions that no one has asked (or answered) before” (M&L p. 12).

Part 5. Study questions with answers

Try answering each question yourself first, then check.

Q1. What are the policies for Biology class?

Source: *Biology Class Basics* slides 3–8. This isn’t in the textbook.

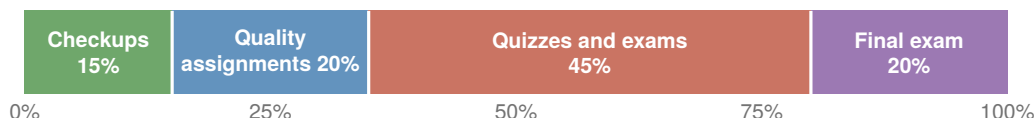
How you know what to do

- **KUDOS** (Know, Understand, Do): a one-page overview of the main topics in the unit.
- **Unit Guide:** more detailed information about what you need to know. It has the important vocabulary, places to find information, and study questions (like this one).
- **Class material:** homework, classwork, lecture notes, and review material. *All* assignments prepare you for the unit exams.
- **Ask a teacher or TA:** if you’re not sure, send an email or come to tutorial/office hours.

Assessments and grades

Type	Examples	Counts toward grade?	Rules
Formative	practice tests, CFAs, homework	No	Practice. A chance for feedback and improvement.
Checkups	cover assigned homework/classwork	Yes, 15%	Not retakeable or made up unless you have an excused absence. Your lowest 2 per semester are dropped .
Quality assignments	—	Yes, 20%	—
Quizzes and exams	mid-unit or end of unit	Yes, 45%	Quizzes are placeholders. The end-of-unit exam can replace your quiz score. You can retake an exam if you score below 80% , but you must show homework completion and complete a retake ticket.

Type	Examples	Counts toward grade?	Rules
Final exam	end of course	Yes, 20%	—



How your grade is weighted (Biology Class Basics, slide 6). Formative work counts 0%, but checkups test it.

Homework

- Expect about **15–30 minutes per night** of *focused* homework (not multitasking).
- Homework isn't checked or graded directly, but the **checkups** assess it. Skipping homework shows up there.
- If you've spent **more than one hour** of focused time and are still stuck, email your teacher and come to the next tutorial.

Academic skills you'll practice all year: note taking, reading comprehension, annotation, writing that is clear and concise, correct academic and scientific vocabulary, CER-style scientific argumentation, and study and test-taking strategies.

Q2. How will students know what they need to learn and how they can demonstrate their understanding in Biology?

Knowing what to learn. The teacher tells you directly, in layers:

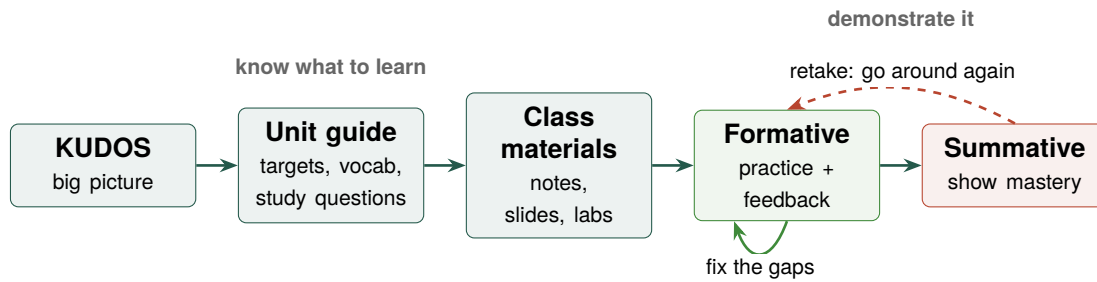
1. The **KUDOS** gives the one-page big picture of the unit: what you should *know*, *understand*, and be able to *do*.
2. The **Unit Guide** breaks the unit into learning targets (1A, 1B, 1C ...). For each one it lists the biology and academic vocabulary, the related classwork (where to find the information), the textbook lessons, and the study questions. If you can answer every study question and use every vocabulary word correctly, you know what the target requires.
3. **Class materials** (notes, slides, labs, homework) teach the content. Every assignment prepares you for the exam.

Demonstrating understanding. You show what you know in two ways:

1. **Formative assessments** (homework, practice tests, CFAs): you practice and get **feedback**. They don't count, so their job is to *reveal gaps* while there's still time to fix them.
2. **Summative assessments** (checkups, quizzes, exams, the final): these measure and grade your understanding.

You demonstrate understanding using the academic skills: accurate vocabulary, clear and concise writing, and especially **CER** (Claim, Evidence, Reasoning). A strong answer makes a claim, supports it with data or observations, and explains *why* the evidence supports the claim.

Connecting it to 1A: the system is built for deliberate practice. Use the unit guide to know the target, practice with formative work, use the feedback to fix gaps, then show mastery on the summative. If you score below 80% on an exam, the retake policy lets you run the loop again.



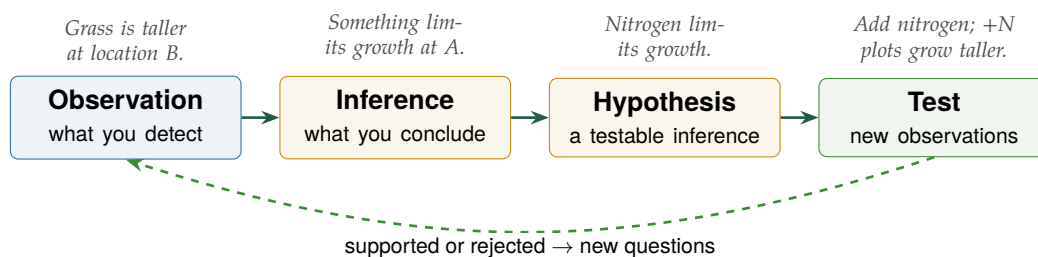
Know the target, practice with feedback, then show what you know. A retake (score below 80%) sends you around the loop again.

Q3. How are observation and inference related to science?

Short answer: Science begins with observation. Inferences are the explanations scientists build from those observations. A good inference becomes a testable hypothesis, and new observations then test it. Science is a cycle of observing and inferring.

In detail:

1. **Observation comes first.** “Scientific investigations begin with observation, the act of noticing and describing events or processes in a careful, orderly way” (M&L p. 12). Scientists observe with their senses and with instruments that extend their senses, like microscopes and thermometers (Campbell p. 16). Recorded observations are **data** (Campbell p. 16), either quantitative or qualitative.
1. **Inference interprets the observation.** “An inference is a logical interpretation based on what scientists already know” (M&L p. 12). An observation tells you *what*. An inference proposes *why* or *what it means*. Inferences use prior knowledge, so the same observation can lead different people to different inferences.
1. **Inference leads to a hypothesis.** “Inference, combined with a creative imagination, can lead to a hypothesis,” a tentative explanation that can be tested (M&L p. 12).
1. **More observations test it.** The hypothesis predicts what *should* be observed if it’s true. Scientists collect new data through experiments or field observations. The data support or reject the inference.



Observation starts the chain; each inference becomes a testable hypothesis, and new observations test it.

Worked example: the salt marsh (M&L Figure 1-2, p. 11)

Step	What happened
Observation	Marsh grass grows taller in some places than others.
Inference	Something is limiting growth in the short-grass places.

Step	What happened
Hypothesis	Marsh grass growth is limited by available nitrogen.
Test (new observations)	Add nitrogen to experimental plots, none to control plots, and measure growth.
Result	Nitrogen plots grew taller → hypothesis supported.

Worked example: Sherlock Holmes (class worksheet). Sherlock notices small details, such as a tan line, a worn sleeve, or mud on a shoe. Those are **observations**, and most of them are **qualitative** (descriptions, not measurements). From them he draws conclusions about where someone has been or what their job is. Those are **inferences**. He is effective because his observations are unusually careful and his inferences use a lot of background knowledge. But an inference can be wrong even if the observation is right. That's why science, unlike a detective story, *tests* inferences.

Worked example: Geotastic (class activity). In the game, the clues you see (language on signs, which side of the road cars drive on, vegetation) are observations. Your guess of the location is an inference. The class prediction was that more “advances” (more observations) would bring guesses closer to the real location, meaning a smaller distance in km. That is the scientific idea that **more and better observations lead to more accurate inferences**.

Why the distinction matters in science:

- Mixing up observation and inference leads to bad conclusions. You might treat a guess as a fact.
- When you write a CER, your **evidence** must be observations/data, and your **claim** is an inference. Keeping them separate is what makes the argument convincing.

Q4. What is the difference between induction and deduction? How does this relate to science?

The difference

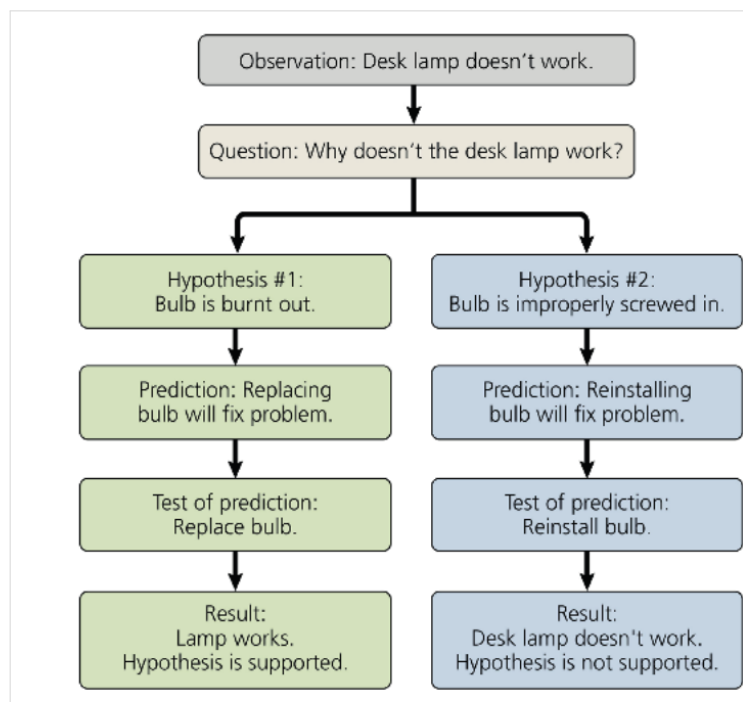
	Induction (inductive reasoning)	Deduction (deductive reasoning)
Direction	Specific → general	General → specific
Starts with	Evidence, data, observations	General rules or premises
Ends with	A general rule or pattern that explains the evidence	A specific conclusion or prediction
Certainty	Not guaranteed. Only <i>probably</i> true, based on past experience.	Guaranteed if the premises are true and the logic is valid.
Teacher's example	“The sun has always risen. Therefore, the sun will rise tomorrow.”	“All dogs have four legs. My pet is a dog. Therefore, my pet has four legs.”
Campbell's example	“All organisms are made of cells” (from two centuries of observations)	“If the burnt-out bulb hypothesis is correct, then the lamp should work if you replace the bulb.”

- **Induction** (Campbell p. 17): “we derive generalizations from a large number of specific observations.” It assumes the past predicts the future, and that isn't always true. So an inductive conclusion is only “probabilistically true based on past experience” (teacher's slide 4).

- **Deduction** (Campbell p. 17): “logic that flows in the opposite direction, from the general to the specific.” The conclusion can only be false if one of the premises is false (teacher’s slide 3).

How it relates to science: science uses both, in a cycle.

1. **Induction builds hypotheses and theories.** Scientists observe many specific cases and look for a pattern. The hypothesis is “a rational accounting for a set of observations, based on the available data and guided by inductive reasoning” (Campbell p. 17).
2. **Deduction makes testable predictions.** From the hypothesis, scientists reason: *if* the hypothesis is true, *then* this specific result should happen. “This deductive testing takes the form of ‘If . . . then’ logic” (Campbell p. 17). Example: *If* nitrogen limits marsh grass growth, *then* grass given extra nitrogen should grow taller than grass without it.
3. **Experiments check the prediction.** If the result matches, the hypothesis is supported. If not, it’s rejected or revised. Then the cycle repeats.



Deduction in action. Each hypothesis leads to an if ... then prediction (“if the bulb is burnt out, replacing it will fix the lamp”), which a test supports or rejects.

Source: Campbell Biology, Figure 1.22, p. 17

Why science never “proves” anything. The teacher’s slide 6 says science “is not a deductive process, it is an inductive one.” The general ideas science rests on come from induction, so they can never be 100% certain:

- More supporting data *increases the probability* that a theory is true, but never makes it certain.
- New data that goes against the pattern can reduce that probability.
- Someone might find a pattern or theory that explains the existing evidence better.

The textbooks say the same thing: “science rarely ‘proves’ anything” (M&L p. 10), and “a hypothesis can never be proved beyond all doubt” (Campbell p. 17). Confidence grows as a hypothesis survives more tests, and this is how scientific consensus forms.

A connection to the Sherlock worksheet (Q16 on that sheet). Sherlock is called the “Master of Deduction,” but most of what he does is **induction**. He starts from specific observations (a tan line, a limp) and reaches a general conclusion (“you’ve been in Afghanistan”). His conclusions are very probable, not guaranteed. So a strong answer is: *No, not really. He mostly reasons from specific observations to a general conclusion, which is inductive reasoning. His conclusions could be wrong if another explanation fits the same clues.* (Arguing the other way is possible when he applies a general rule to a specific case, for example “people who have been in the tropics are tanned; this man is tanned ...”. Either way, name the direction of reasoning and use a specific example from the clip.)

Sources

- Miller, K. R., & Levine, J. S. *Miller & Levine Biology*. Pearson, 2019. Lesson 1.1 (pp. 8–14), Lesson 1.2 (pp. 15–18).
- Urry, L. A., et al. *Campbell Biology*. Ch. 1, Concept 1.3 (pp. 16–17).
- Teacher materials: *Ecology Unit Guide (KJ)*, *Biology Class Basics (KJ_PH)*, *Deductive vs Inductive Reasoning Overview*, *Observation and Inference*, *Geotastic Inference Game*.